What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Antivirals 1 of 2: Drugs to Treat Herpesvirus Infections

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Adapted by: Trager Hintze, Diptiman Bose, Ray Brown
Editors: Jordan Finn, Russell Johnson

Adapted from: Drugs to Treat Herpesvirus Infections

Listen to this Brick
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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action and mechanism of resistance of antiviral agents used in the treatment of herpesvirus infections.
2. List the indications for antiviral agents used in the treatment of herpesvirus infections.
3. Recognize th… d with drugs used in the treatment of herpesvirus infections.
4. Identify the common drug interactions associated with drugs used in the treatment of herpesvirus infections.

CASE CONNECTION

SL is at the university student health center for an acute problem with burning pain when urinating. "It just started yesterday, and it's getting worse. It even hurts when I'm not urinating," she reports. SL has never had this before. "The pain is more to one side. I think there is a sore on the skin there, and it feels swollen. And I've been feeling ill, kind of achy and maybe like I had a mild fever." Exam reveals multiple ulcerative lesions on an erythematous base around the genital area.

How will you diagnose and treat SL's condition? Consider your answer as you read, and we'll revisit SL at the end of the brick.

GO TO CONCLUSION ↓

What Are Antiherpetic Drugs?

From cold sores to genital herpes, we collectively dread the painful and sometimes embarrassing symptoms herpesviruses can cause. Although they are incredibly contagious, herpesviruses can be managed with antiherpetic drugs. They can't be cured, but the hassle they cause can be greatly reduced.

How Do Antihe

Viruses require the host cell organelles and metabolic resources to replicate. As a result, antiviral medications are geared toward preventing the virus from completing its metabolic process for self-replication. Antiviral medications work in a variety of ways and are classified by their mechanism of action. Below is a summary of antiviral medications that are used to treat Herpes viruses and their effect on the viral life cycle (Figure 1). These medications can be used for a variety of viral infections.

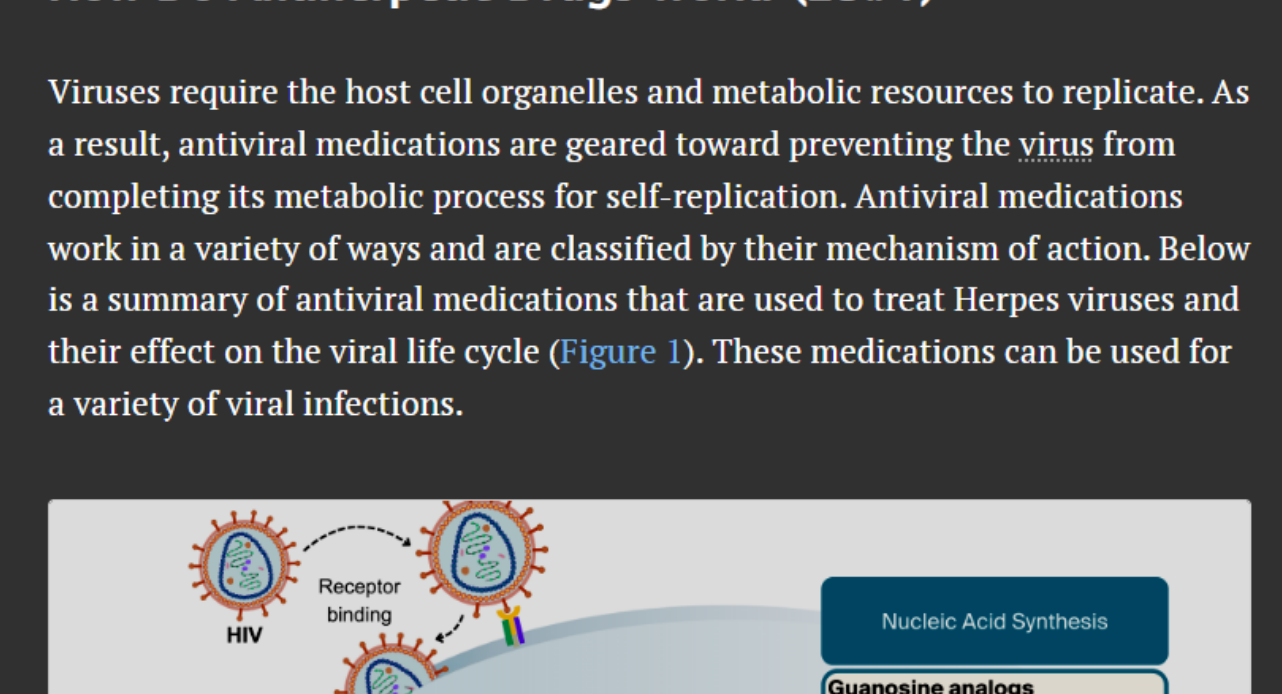


Figure 1

Nucleic Acid Synthesis Inhibitors

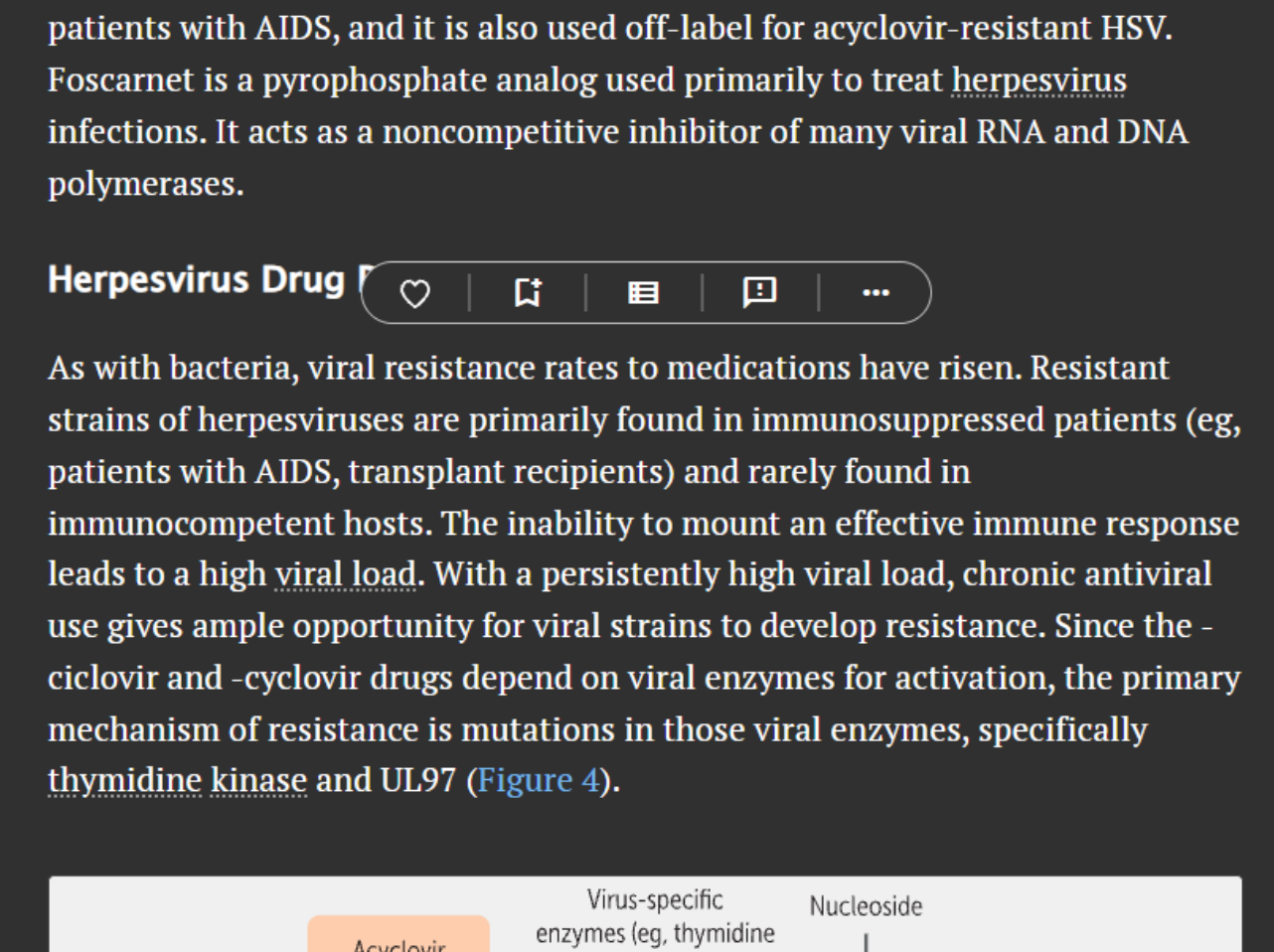
Nucleic acid synthesi… lasses that all inhibit DNA and RNA synthesis of viruses. Each class has a unique mechanism for blocking the nucleic acid synthesis pathway.

Guanosine Analogs. These drugs have the suffix -cyclovir or -ciclovir. Included in this class are **acyclovir**, **valacyclovir**, **famiciclovir**, and **ganciclovir**, and these drugs are specific for herpes viruses because they are activated by viral kinases. Guanosine analogs consist of guanosine, a purine, attached to a sugar. Their chemical structure mimics nucleosides and can become incorporated into growing DNA strands. When added to DNA, their chemical structure prevents lengthening of the DNA strand and thereby reduces viral proliferation. Because they are not specific to viral DNA, they also affect mitochondrial DNA and have side effects. *Acyclovir, valacyclovir, and famciclovir are used for HSV-1, HSV-2, and varicella-zoster virus* (VZV; HHV-3). *Ganciclovir is used for cytomegalovirus* (CMV; HHV-5) and *acyclovir-resistant HSV.*

Viral DNA Polymerase Inhibitors. This class of antivirals acts on the viral DNA polymerase by imitating nucleotides. Viral DNA polymerase has structural differences from the human DNA polymerase, and this allows for quasi-selective toxicity (the selective toxicity is not fully selective), but there still may be some interference with human DNA polymerase. **Cidofovir** and **foscarnet** are the two primary medications in this class. Both are primarily used in immunocompromised patients. Cidofovir is used for CMV retinitis in patients with AIDS, and it is also used off-label for acyclovir-resistant HSV. Foscarnet is a pyrophosphate analog used primarily to treat herpesvirus infections. It acts as a noncompetitive inhibitor of many viral RNA and DNA polymerases.

Herpesvirus Drug

As with bacteria, viral resistance rates to medications have risen. Resistant strains of herpesviruses are primarily found in immunosuppressed patients (eg, patients with AIDS, transplant recipients) and rarely found in immunocompetent hosts. The inability to mount an effective immune response leads to a high viral load. With a persistently high viral load, chronic antiviral use gives ample opportunity for viral strains to develop resistance. Since the -ciclovir and -cyclovir drugs depend on viral enzymes for activation, the primary mechanism of resistance is mutations in those viral enzymes, specifically thymidine kinase and UL97 (Figure 4).


Acyclovir
Penciclovir
Ganciclovir
Drug resistance, mutations
Virus-specific enzymes (eg, thymidine kinase, UL97)
Nucleoside
Monophosphate
Trifluridine
Cidofovir
Host kinases
Diphosphate
Foscarnet
Triphosphate
Incorporation into viral DNA
Competitive inhibition of viral DNA polymerase
termination
DNA synthesis

QUIZ

Figure 4

Thymidine Kinase Mutation. In the case of herpes simplex, drug resistance to acyclovir and related drugs has been linked to mutations, resulting in decreased viral thymidine kinase activity. Thymidine kinase phosphorylation is the first step of activating these drugs, so decreased activity results in decreased availability of the active drug. Drugs in the family of acyclovir all have the same mechanism of action; therefore, the thymidine kinase mutation confers resistance to all of these drugs. In cases of resistant herpes simplex infection, foscarnet may be used because it inhibits the viral DNA polymerase without need for activation.

***UL97* Mutation.** In the case of CMV, ganciclovir resistance may be a result of *UL97* gene mutation. The UL97 phosphotransferase affects the initial phosphorylation of ganciclovir needed for drug activation. There is no clear guideline for treating ganciclovir-resistant CMV, but some have suggested adding foscarnet to ganciclovir or increasing the dose of ganciclovir.

What suffix is common to guanosine analog drugs?

What are the indications for antiherpetic drugs? (LO#2)

Herpes simplex 1 (HSV-1) and herpes simplex 2 (HSV-2) viruses are part of the Herpesviridae family and are responsible for orolabial and genital herpes, respectively. After primary infection, the virus lays dormant in the cell bodies of neurons and may reactivate in the future when the immunity wanes. HSV-1 and HSV-2 may be treated with **acyclovir, valacyclovir, famciclovir, and ganciclovir**. Famciclovir and valacyclovir are unique in that they are prodrugs. Prodrugs are metabolized into their active form within the host. Famciclovir is a prodrug with better gastrointestinal (GI) absorption. Valacyclovir is a prodrug of acyclovir with better GI absorption and bioavailability.

Localized Infection. Localized infection can be treated topically or orally. Topical treatment is usually reserved for mild cases because it is less effective than oral treatment. Oral medication is used for treating primary infection and recurrent breakouts of oral and genital herpes. Intravenous (IV) medication is reserved for the most severe cases:

- Penciclovir is a topical treatment for orolabial herpes only.
- Valacyclovir and
- Acyclovir is the only drug of this class that can be used orally and IV.

Varicella-Zoster Virus

Varicella-zoster virus (VZV), or HHV3, is responsible for chickenpox and shingles. As with herpes simplex, the virus lays dormant in the cell bodies of neurons. Primary infection results in fever, headache, and diffuse pustular rash. Shingles are a recurrent breakout rash occurring after primary infection with VZV. It is most commonly seen in the elderly but may also occur in immunosuppressed patients. Both chickenpox and shingles may be treated with oral valacyclovir or acyclovir, however, most often chickenpox is not treated.

CLINICAL CORRELATION

The varicella vaccine was introduced in the US in 1995. That's why many kids don't get chicken pox as a child. However, with some waning vaccination rates in the US there are many more cases of chicken pox. Why don't we give an antiviral medication, like acyclovir, to them? As you have learned the virus is self-limiting. The red itchy 'pox' are bothersome, but will go away. The antiviral medication might only reduce symptoms for a day or two, if they start taking it early enough, and it makes the patient susceptible to a variety of adverse effects that we will discuss below. Ultimately, the risk outweighs the benefit.

Epstein-Barr Virus

Acute infection with … uses infectious mononucleosis. In immunocompetent hosts, infection is subclinical or may result in a flu-like viral syndrome. In immunocompromised patients, symptoms may be severe due to acute infection or latent virus from a past infection reactivating. Complications may include splenomegaly and upper airway obstruction from pharyngeal swelling. Infectious mononucleosis is primarily managed with symptomatic treatment. Acetaminophen and nonsteroidal anti-inflammatory drugs are used for fever, sore throat, and malaise. Antiviral medications have not been shown to be clinically effective against symptoms or duration of EBV infection.

Cytomegalovirus

Cytomegalovirus (CMV), or HHV5, is transmitted through close contact, sexual contact, blood, and tissue exposure. Primary CMV infection in the immunocompetent host typically ranges from being asymptomatic to causing mononucleosis. If symptomatic, it is self-limited and resolves within weeks. Antiviral therapy has not proven effective in the immunocompetent patient due to the high rates of self-resolution. Treatment for these patients focuses on symptomatic relief.

For the immunocompromised, complications of infection may be severe. Antiviral treatment is given when symptomatic or for prophylaxis if the patient is at increased risk of complications (eg, transplant recipients). Antiviral medications given for CMV include **ganciclovir, valganciclovir, cidofovir, and foscarnet**. Ganciclovir and valganciclovir (prodrug of ganciclovir) work similarly to acyclovir in that they rely on the viral thymidine kinase for activation. Valganciclovir is given orally; ganciclovir is IV. In the setting of CMV, foscarnet and cidofovir are used for CMV retinitis in patients with HIV/AIDS. Both drug

What are the primary antiviral medications used for CMV in immunocompromised patients?

What Are Some Adverse Effects of Antiherpetic Drugs? (LO#3)

Antiherpetic drugs are generally well tolerated. One common adverse reaction among all antiherpetic drugs is the potential for renal impairment. Some degree of renal impairment occurs in most patients on foscarnet, so renal function must be monitored during use. This can also be managed with aggressive pre-hydration with fluids. Renal impairment with IV acyclovir and ganciclovir occurs due to crystal nephropathy, which is precipitation in the renal tubules of crystalized drug. This is avoided by maintaining proper hydration. Other common adverse effects include:

- Foscarnet: Renal … ces including hypokalemia, hyp…
- Acyclovir: Increased liver enzymes and increased seizure risk. Phlebitis (vein irritation) if administered IV.
- Ganciclovir: Leukopenia, thrombocytopenia, and anemia.

What are common adverse effects of ganciclovir?

What are the drug interactions associated with antiherpetic drugs? (LO#4)

Antiviral medications, like medications in other classes, can have interactions with other drugs. As patients accumulate multiple medications in their regimen, their risk for drug interaction increases. This is very true for patients who are immunocompromised or have chronic illnesses. Since all of the antiherpetic medications are renally cleared, any other medication that can be nephrotoxic, or patie… be avoided, or used cautiously.

CLINICAL CORRELATION

A common drug-drug interaction you might encounter is when *ganciclovir* is used to treat CMV in the immunocompromised HIV patient on the medication *tenofovir.* These medications directly compete for excretion in the kidneys, as such drug levels increase, and it can cause severe renal damage. This combination should be avoided.

Why should ganciclovir and tenofovir be avoided together?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to SL, … ition?

SL's symptoms and physical findings are consistent with genital herpes infection, which is caused by HSV-2. You explain, "It looks like you have a herpes infection. I know that sounds serious; it can be. But it is good that you came in quickly. I will send a test to confirm the diagnosis, and meanwhile, we can get you started on acyclovir to start to treat it." SL is upset to hear this bad news, and you answer her questions about the treatment and prognosis. You discuss possible adverse effects and tests you may need to get. You also discuss the need for testing for other sexually transmitted infections and related issues.

Summary

- Antiherpetic drugs manage symptoms of herpesviruses but do not cure them.
- Guanosine analogs, such as acyclovir and valacyclovir, inhibit viral DNA synthesis and are activated by viral kinases.
- Viral DNA polymerase inhibitors like cidofovir and foscarnet are used primarily in immunocompromised patients.
- Herpesvirus drug resistance often occurs in immunosuppressed patients due to mutations in viral enzymes.
- Common adverse effects of antiherpetic drugs include renal impairment and, for ganciclovir, leukopenia, thrombocytopenia, and anemia.
- Drug interactions, particularly with nephrotoxic medications, must be carefully managed to avoid severe renal damage.

Review Questio

1. A 73-year-old patient with a history of a kidney transplant is hospitalized for pneumonia. During his hospitalization, a vesicular rash is noted on his trunk that does not cross the midline. He is started on acyclovir. What is the mechanism of action of acyclovir?

A. ● Inhibition of viral DNA synthesis by competing with deoxyguanosine triphosphate for incorporation into viral DNA
B. ○ Inhibition of viral neuraminidase
C. ○ Inhibition of viral RNA–dependent DNA polymerase
D. ○ Inhibition of viral RNA–dependent RNA polymerase and action as a chain terminator
E. ○ Inhibition of viral thymidine kinase

Hide Explanation

The correct answer is inhibition of viral DNA synthesis by competing with deoxyguanosine triphosphate for incorporation into viral DNA (A). Acyclovir is used to treat herpes simplex 1 and 2. Inhibition of viral DNA synthesis results in reduced viral replication and may shorten the duration and severity of viral reactivation. This patient is immunosuppressed from the chronic medications transplant recipients use. This puts him at increased risk of reactivation of dormant viral infection. The presentation of his rash is characteristic of shingles, the reactivation of varicella-zoster virus (VZV). Acyclovir is the appropriate treatment for both primary infection and reactivation of infection with VZV, HSV-1, and HSV-2. Inhibition of viral neuraminidase (B) is the mechanism of action of oseltamivir, which is used for influenza. Nucleosid… n the treatment of HIV infection and compete for incorporation into viral DNA by their inhibition of viral RNA–dependent DNA polymerase (C). Inhibition of viral RNA–dependent RNA polymerase and action as a chain terminator (D) is the mechanism of action of sofosbuvir. Inhibition of viral thymidine kinase (E), which is one of the multiple enzymes that help convert acyclovir into its active form, is not a mechanism of action of acyclovir.

2. A 19-year-old college student comes into the clinic complaining of fever, sore throat, chills, and fatigue for 3 days. Physical exam shows mild tenderness in the left upper abdomen with palpation. The clinician orders tests for viral causes of upper respiratory infection. Which of the following viruses have no established antiviral treatment?

A. ○ Cytomegalovirus
B. ● Epstein-Barr virus
C. ○ Hepatitis B virus
D. ○ Hepatitis C virus
E. ○ Varicella-zoster virus

Hide Explanation

The correct answer is Epstein-Barr virus (B). EBV is responsible for infectious mononucleosis. Curr… en to change outcomes in patients with symptomatic EBV infection, so patients are managed symptomatically. The patient has infectious mononucleosis, which causes upper respiratory symptoms, fever, and occasionally causes splenomegaly, which leads to abdominal pain. EBV infection may be diagnosed with a blood test. Cytomegalovirus (A) is usually treated symptomatically but may be treated with antivirals such as ganciclovir when a patient is immunocompromised. Hepatitis B virus (C) may be treated with antiviral nucleoside/nucleotide analogs. Hepatitis C virus (D) may be treated with direct-acting antivirals. Varicella-zoster virus (E) primary infection and reactivation are both treated with acyclovir.

3. A 45-year-old male with a diagnosis of HIV has been recently diagnosed with a CMV infection. He has a known history of drug-resistant CMV. On exam, the patient is found to have CMV retinitis, foscarnet treatment is initiated. Which of the following is the most concerning side effect?

A. ○ Elevated liver enzymes
B. ○ Liver impairment
C. ● Renal impairment
D. ○ Thrombocytopenia

Hide Explanation

The correct answer is renal impairment (C). Foscarnet readily impairs renal function, and can cause signif… e effect aggressive fluid pre-hydration is employed. Elevated liver enzymes (A) and liver impairment (B) are NOT adverse effects of foscarnet, these effects are more associated with acyclovir treatment. Lastly, thrombocytopenia (D) is incorrect as this is one of the myelosuppressive effects of ganciclovir.

4. A 38-year-old female with a diagnosis of HIV has been recently diagnosed with a CMV infection. Part of her HIV medication regimen includes tenofovir. Which medication to treat CMV should be avoided in this patient?

A. ○ Acyclovir
B. ○ Famciclovir
C. ○ Foscarnet
D. ● Ganciclovir

Hide Explanation

The correct answer is ganciclovir (D). Ganciclovir and tenofovir compete for excretion in the kidneys. This can cause an increase in their levels, causing severe renal damage. This combination of medications should be avoided. Acyclovir (A) and famciclovir (B) do not treat CMV, nor do they have the same interaction with tenofovir. Foscarnet (C) can cause significant renal impairment, however, it does not have a drug interaction with tenofovir.

and famciclovir (B) do not treat CMV, nor do they have the same interaction with tenofovir. Foscarnet (C) can cause significant renal impairment, however, it does not have a drug interaction with tenofovir.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action and mechanism of resistance of antiviral agents used in the treatment of herpesvirus infections.~~
02 ☑ ~~List the indications for antiviral agents used in the treatment of herpesvirus infections.~~
03 ☑ ~~Recognize the common adverse reactions associated with drugs used in the treatment of herpesvirus infections.~~
04 ☑ ~~Identify the common drug interactions associated with drugs used in the treatment of herpesvirus infections.~~

Objective Confidence: 4/4